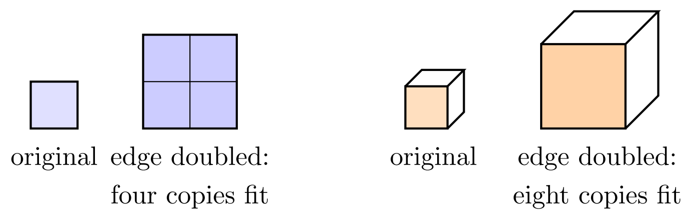
Scale Factor Confusion: Linear, Square, Cubic

High School Geometry

Scaling an area by the square of the factor, scaling a volume by its cube, and recovering the linear factor from an area or volume ratio

Directions:

- For similar figures with linear scale factor k: every *length* scales by k, every *area* (including surface area) scales by k^2, and every *volume* scales by k^3. The exponent counts the dimensions being scaled.
- Going backwards, take the matching root: $k = \sqrt{\text{area ratio}}$ and $k = \sqrt[3]{\text{volume ratio}}$.
- State which rule you are using before you compute. On a **find the mistake** problem, name the error in words *and* show correct work.



Why the exponent matches the dimensions. No measurements are shown here — use the numbers given in each problem.

1. A polygon has an area of 12 cm^2. It is dilated by a scale factor of 3. Find the area of the image.

Answer: ________________ cm^2

2. Two cones are similar. The smaller one has a volume of 54 cm^3. Every linear measurement of the larger cone is 2 times the matching measurement of the smaller. Find the volume of the larger cone.

Answer: ________________ cm^3

3. **Find the mistake.** Two triangles are similar with linear scale factor $5 : 2$ (larger to smaller). The smaller triangle has area 18 cm^2. Rey writes:

$$A_{\text{larger}} = 18 \cdot \frac{5}{2} = 45 \text{ cm}^2$$

(a) Name Rey's error in words. (b) Show correct work and give the larger area.

Answer: ________________ cm^2

4. Two similar pentagons have areas in the ratio 25 : 9. The linear scale factor k therefore satisfies

$$9k^2 = 25$$

(a) Solve the equation for *every* value of k. (b) Which value can be a scale factor for a pair of similar figures, and why must the other be rejected?

Answer: ______________

5. Find the mistake. Two cylinders are similar. The smaller has volume 250 cm^3; the larger has volume 2000 cm^3. Two students disagree:

Nia: "The scale factor is $2000/250 = 8$." *Kim:* "No, it is $\sqrt{8} \approx 2.83$."

Both are wrong. (a) Name the rule each student misapplied. (b) Give the correct linear scale factor.

Answer: ______________

6. A classmate claims: *"If I double every edge of a rectangular storage box, I will need twice as much paint for the outside, and it will hold twice as much sand."*

Using a labelled sketch and the scale-factor rules, explain what really happens to the paint and to the sand, and say why the exponent matches the number of dimensions being scaled.